

ADVANCED ASSIGNMENT | W2W3_JUN26

Key Drivers of Yearly Spending TokoBli E-Commerce Customers

Correlation and Stepwise Regression Analysis of 299 Customers

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[W2 JUN26 UBP] Reinaldy Zulfananda Arkaan Intermediate

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Business Context and Analytical Approach

The Product Manager wants to know which factors really drive customer yearly spending

Business Question

Which factors have a significant effect on Yearly Spending, and how much would a new customer potentially spend?

Data Used

299 TokoBli customers with 6 independent variables (Age, Income, Tenure, Avg. Session Time, Total Promo, Bounce Rate) and 1 dependent variable (Yearly Spending)

Analysis Workflow

1

Correlation

Measure how closely variables relate

2

Regression

Measure each effect on Yearly Spending

3

Stepwise

Filter variables until all are significant

4

Prediction

Estimate spending for a new customer

Data Overview

No missing values across the 299 rows of data



Variable Notation Used Throughout the Analysis

X1	Age	Customer age
X2	Income	Customer income per year
X3	Tenure	Length of time as a customer
X4	Avg. Session Time	Average session duration on the platform
X5	Total Promo	Number of promos the customer received
X6	Bounce Rate	Share of visits with no follow-up action
Y	Yearly Spending	Total customer spending per year (target)

Case 1 | Correlation Calculation

Pearson correlation across all variables, computed with the CORREL function in the spreadsheet

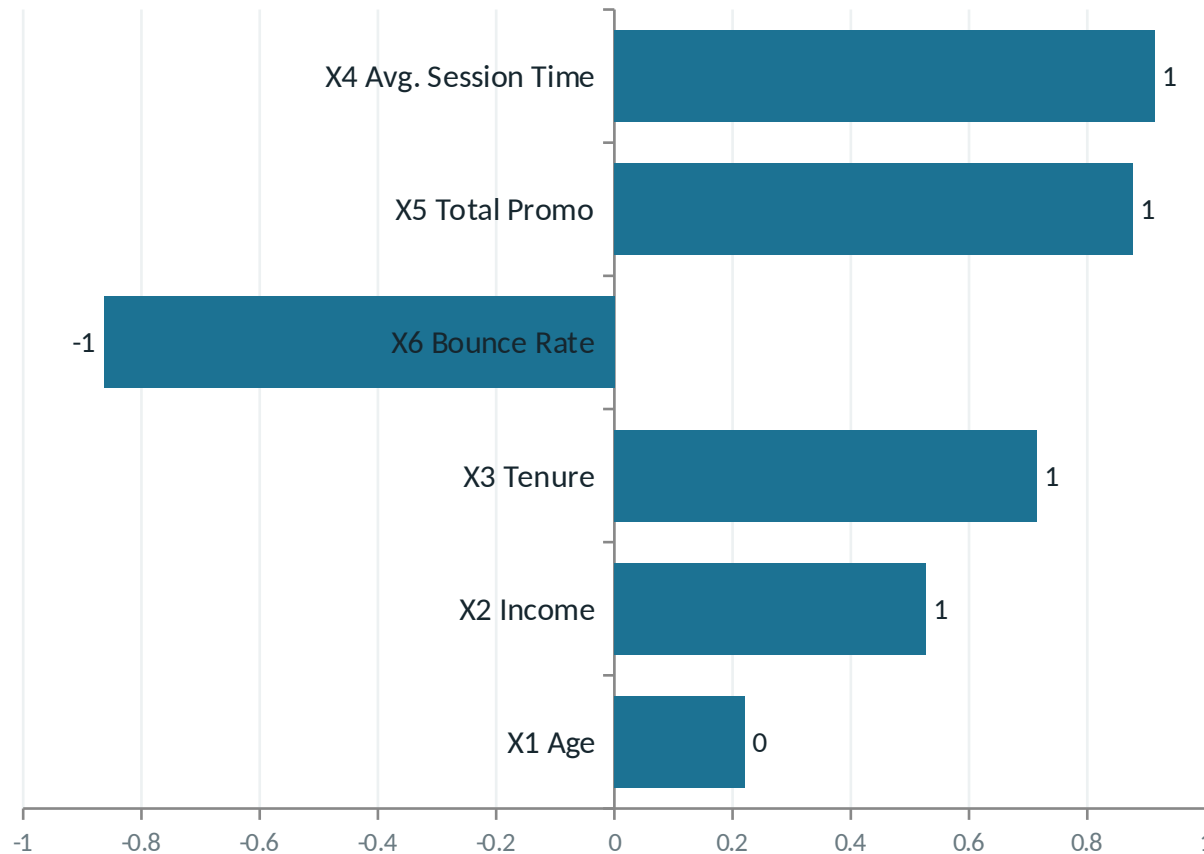
	X1	X2	X3	X4	X5	X6	Y
X1 Age	1.000	0.141	0.160	0.193	0.208	-0.129	0.221
X2 Income	0.141	1.000	0.379	0.484	0.446	-0.448	0.527
X3 Tenure	0.160	0.379	1.000	0.685	0.621	-0.570	0.715
X4 Avg. Session Time	0.193	0.484	0.685	1.000	0.777	-0.787	0.914
X5 Total Promo	0.208	0.446	0.621	0.777	1.000	-0.794	0.877
X6 Bounce Rate	-0.129	-0.448	-0.570	-0.787	-0.794	1.000	-0.863
Y Yearly Spending	0.221	0.527	0.715	0.914	0.877	-0.863	1.000

Colored cells mark strong correlations ($|r|$ above 0.5). Green means same direction, red means opposite direction.

The three variables most strongly related to Yearly Spending: Avg. Session Time (0.914), Total Promo (0.877), and Bounce Rate (-0.863).

Case 1 | Insights from the Correlation

How strongly each variable relates to Yearly Spending



Avg. Session Time matters most

Correlation of 0.914. The longer a customer engages with the platform, the higher the yearly spending.



Promos show a strong relationship

Correlation of 0.877 between the number of promos received and the amount spent.



Bounce Rate holds spending down

Correlation of -0.863. The more often visits end without action, the lower the yearly spending.



Age is not a differentiator

A correlation of only 0.221, so age-based segmentation is far less relevant.

Grading: show the insights

From Correlation to Regression

Correlation answers how close, regression answers how much

Correlation does not measure impact size

The r value only shows direction and strength, not how many rupiah spending rises for one unit of change.

Correlation evaluates one pair at a time

Multiple regression evaluates all variables at once, so the pure effect of each one becomes visible.

There are signs of multicollinearity

X4, X5, and X6 are highly correlated with each other (up to -0.794), so they need to be tested through regression and VIF.

A model is needed for prediction

Predicting a new customer spending level is only possible with a regression equation, not with correlation.

Grading: escalate correlation into regression

Case 2 | Multiple Regression - Iteration 1

$Y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_5x_5 + b_6x_6$ with alpha 0.05

Variable	Coefficients	P-value	Decision	Conclusion
Intercept	1,947,880	0.0012	H1 accepted	Significant
X1 Age	30,555	0.0688	H0 accepted	Not Significant
X2 Income	0.0358	0.0014	H1 accepted	Significant
X3 Tenure	64,158	0.0000	H1 accepted	Significant
X4 Avg. Session Time	6,287	0.0000	H1 accepted	Significant
X5 Total Promo	133,216	0.0000	H1 accepted	Significant
X6 Bounce Rate	-50,350	0.0000	H1 accepted	Significant

Goodness of Fit

R Square
0.9293
Adjusted R Square
0.9278
Significance F
close to 0

Variable eliminated

X1 Age

Its p-value of 0.0688 is above 0.05 and the largest of all, so it is dropped in the next iteration.

Case 2 | Stepwise Regression - Iteration 2

Backward elimination: the model is recalculated without X1 Age

Variable	Coefficients	P-value	Decision	Conclusion
Intercept	2,590,241	0.0000	H1 accepted	Significant
X2 Income	0.0368	0.0011	H1 accepted	Significant
X3 Tenure	64,576	0.0000	H1 accepted	Significant
X4 Avg. Session Time	6,342	0.0000	H1 accepted	Significant
X5 Total Promo	136,148	0.0000	H1 accepted	Significant
X6 Bounce Rate	-49,209	0.0000	H1 accepted	Significant

All p-values are below 0.05

The elimination process stops here.
Two iterations in total, with one variable removed.

Final Model

$$Y = 2,590,241 + 0.0368 x_2 + 64,576 x_3 + 6,342 x_4 + 136,148 x_5 - 49,209 x_6$$

Adjusted R Square of 0.927, so the model explains about 92.7 percent of the variation in Yearly Spending.

Multicollinearity Check

Highest correlation among X
-0.794 (limit 0.8)

Highest VIF
3.78 on X4 (limit 5)

Conclusion: the model is safe, no further variables need to be removed.

Case 3 | Predicting a New Customer

Profile of a customer who joined about one month ago

Variable	Value	Coefficient	Contribution
X1 Age	20	not used	0
X2 Income	20,000,000	0.0368	736,863
X3 Tenure	1	64,576	64,576
X4 Avg. Session Time	1,000	6,342	6,341,902
X5 Total Promo	20	136,148	2,722,954
X6 Bounce Rate	30	-49,209	-1,476,274
b0 Intercept		2,590,241	2,590,241

X1 Age is still shown for completeness, but it is not counted because it is not significant in the final model.

Potential Yearly Spending

Rp10,980,264

About Rp11 million per year

13 percent above the average of all customers (Rp9,700,334)

How to read it

The largest contributions come from Avg. Session Time (Rp6.3 million) and Total Promo (Rp2.7 million), while Bounce Rate holds back about Rp1.5 million.

Grading: predict potential yearly spending

Case 4 | Key Insights

A summary of the findings from correlation, regression, and prediction

Rp6,342

per 1 unit of Avg. Session Time

Session duration is the strongest driver of yearly spending, both in the correlation (0.914) and in the regression.

Rp136,148

per 1 additional promo

Promos deliver the largest per-unit impact on the rise in customer yearly spending.

-Rp49,209

per 1 point of Bounce Rate

An experience that makes visitors leave without acting eats directly into yearly spending.

92.7%

of Y variation explained

Five behavioral and financial variables are enough to explain almost all of the variation in spending.

Age is not significant, so the strategy should target customer behavior rather than age demographics.

Case 4 | Recommendations for the Product Manager

Four priorities ranked by the size of their impact on Yearly Spending

1

Make Avg. Session Time the primary KPI

Personalize product recommendations, improve search relevance, and add review content so customers stay longer on the platform.

Highest impact

2

Broaden promo reach and targeting

Apply bundling and tiered vouchers based on purchase history, since one extra promo is worth around Rp136 thousand.

Largest per-unit impact

3

Push Bounce Rate down

Speed up page loading, improve campaign landing pages, and simplify the checkout flow so visits end in transactions.

Experience improvement

4

Build a tenure-based loyalty program

Offer tiered benefits by length of membership to lift retention, since tenure is proven to raise spending significantly.

Medium term

Grading: insight and recommendation

Conclusion

- Five variables have a significant effect on Yearly Spending: Income, Tenure, Avg. Session Time, Total Promo, and Bounce Rate.
- The final stepwise model (2 iterations, 1 elimination) explains 92.7 percent of the variation in spending and passes the multicollinearity check.
- The new customer in the case is predicted to spend about Rp10.98 million per year, above the TokoBli customer average.
- Campaign strategy should focus on customer behavior, especially session duration, promos, and lowering bounce rate.

Thank You

URL :

<https://docs.google.com/spreadsheets/d/19ewLgDPTgC3NhGHpAa-ISrVMpCnA21sae1fpixEBgx8/edit?usp=sharing>